

## APPENDIX

### A. Proof of Proposition 1

*Proof Sketch.* The sum of distances is easy to check in polynomial time, i.e., the problem is in NP. To prove the NP-Hardness, we show a reduction from the decision version of the Traveling Salesman Problem (TSP), one of Karp's 21 NP-Complete problems [49].

We first introduce the decision version of the TSP problem: Given an undirected weighted graph  $G = (V, E)$  with  $n$  vertices and their pairwise distances, the problem is to decide whether there exists a path of total weight less than or equal to  $L$ , which visits each vertex exactly once. In addition, if all the vertices are in a plane (i.e., the distances satisfy the triangle inequality), existing studies also prove that the TSP problem is still NP-Complete under  $l_1$  and  $l_2$  distances [50]–[52].

Given an instance of the TSP problem in a plane, we then construct an instance of the Constant Pattern Ordering Problem with 2 dimensions: For each vertex  $v_i \in V$ , we use  $(v_i[1], v_i[2])$  to denote its coordinates, and then construct a tuple  $y_i = (v_i[1], v_i[2]) \in Y$  corresponding to  $v_i$ . Therefore, the distances between vertices  $v_i, v_j$  are also equal to the distances between tuples  $y_i, y_j$ .

Following Problem 2, the decision version of our problem is to decide if there exists a constant pattern order with the sum of the distances less than or equal to  $K$ . Let  $K = L$ , we prove that these two instances are equivalent. Note that the distances of two instances are equal, to find a path of total weight less than or equal to  $L$  in TSP problem, we then order the tuples following the order of the vertices in the path. Therefore, the path in TSP has total weight less than or equal to  $L$ , if and only if there exists an order of tuples with the sum of the distances less than or equal to  $K$ . To conclude, the NP-Completeness of the decision problem is proved.  $\square$

### B. Proof of Lemma 2

*Proof.* First, we consider the distance between  $x$  and  $y$  in the  $j$ -th dimension. When  $Y_i^{\min}[j] < x[j] < Y_i^{\max}[j]$ ,  $x[j]$  is covered by the range of  $Y_i[j]$ . Thereby,  $\theta_j(x, Y_i) = 0$  in this case. When  $x[j] < Y_i^{\min}[j]$ ,  $x[j]$  is out of the value range of the  $j$ -th dimension. Then the value closest to  $x[j]$  is  $Y_i^{\min}[j]$ , which gives the minimum distance  $|x[j] - Y_i^{\min}[j]|$ . Similarly, we could get the distance bound of the  $j$ -th dimension when  $x[j] > Y_i^{\max}[j]$ . In summary, we have

$$|x[j] - y[j]| \geq \theta_j(x, Y_i), \quad \forall y \in Y_i, \quad 0 \leq j \leq l$$

The distance  $\delta(x, y)$  is thus bounded by  $\theta(x, Y_i)$ , for  $y \in Y_i$ .

$$\delta(x, y) = \sum_{j=1}^l |x[j] - y[j]| \geq \sum_{j=1}^l \theta_j(x, Y_i) = \theta(x, Y_i)$$

$\square$

### C. Proof of Proposition 3

*Proof.* Suppose that there exists a neighbor  $y^*$  of  $x$  in cluster  $Y_i$ , i.e.,  $\delta(x, y^*) \leq \delta_\kappa$ . On the other hand, according to the Lemma 2,  $\forall y \in Y_i, \delta(x, y) \geq \theta(x, Y_i)$ . It follows  $\delta(x, y^*) \geq \theta(x, Y_i)$ . We thus obtain  $\theta(x, Y_i) \leq \delta_\kappa$ .  $\square$

### D. Proof of Proposition 4

*Proof.* The proposition can be proved similarly following the proof of Proposition 3.  $\square$

### E. Proof of Proposition 5

*Proof.* Note that the smoothing tuples in our algorithm are selected from the valid modification set  $V_i$ . Thereby, the modification  $x'_i$  of  $x_i$  exists if and only if  $V_i$  is not empty. We discuss two cases of window  $W_i$ .

If  $W_i = \emptyset$ , the smoothness constraint  $\mathcal{S}$  of  $x_i$  is naturally satisfied. Therefore, a candidate can always be found from  $\text{Nei}(x_i)$  in constant patterns  $Y$  to modify  $x_i$  safely.

If  $W_i \neq \emptyset$ , then the candidate modification set  $C_i$  involves  $\text{Nei}(x'_j)$ , for  $x'_j \in W_i$ , i.e.,  $x'_j \in C_i$  as well. Due to the streaming computation,  $x_j$  is smoothed before  $x_i$ . That is,  $x'_j$  in the window satisfies the smoothness constraint with each other, referring to the previous smoothing steps. Therefore, there always exists  $x'_j \in W_i$  as the valid modification of  $x_i$ .  $\square$

### F. Proof of Proposition 6

*Proof.* For window size  $\omega = 0$ , the temporal smoothness of the time series is no longer considered. The window  $W_i$  of each tuple  $x_i$  is always empty and the candidate set  $C_i$  is  $\text{Nei}(x_i)$ . Since the modifications no longer affect each other, we have  $V_i = C_i$ . The one with the minimum distance to  $x_i$  in  $V_i$  returned by Algorithm 2 is indeed the optimal solution.  $\square$